Bubble Radius Fitting

User Guide

Version 1.0

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1 Introduction

Bubble Radius Fitting is a desktop application for extracting bubble radius versus time data from high-speed camera image sequences. The software detects bubble boundaries via adaptive thresholding and morphological filtering, then fits circles using the Taubin algebraic method.

1.1 Key Features

- Three analysis modes: Pre-tune, Manual, and Automatic
- Side-by-side image viewer with zoom, pan, and overlay visualisations
- Interactive ROI (Region of Interest) selection
- Advanced image filters: Gaussian Blur, CLAHE, Morphological Open/Close
- Live-updating $R(t)$ scatter chart
- Data export to `.mat` files (MATLAB-compatible)
- Saveable/loadable parameter presets (JSON)
- Radius outlier filtering

1.2 Supported Image Formats

The application supports standard image file formats commonly produced by high-speed cameras:

Format	**Extensions**
TIFF (preferred)	`.tiff`, `.tif`
PNG	`.png`
JPEG	`.jpg`, `.jpeg`
BMP	`.bmp`

When multiple formats are present in the same folder, TIFF files are given priority. Files within each format are sorted using natural ordering (e.g. `img1`, `img2`, `img10` rather than `img1`, `img10`, `img2`).

Both 8-bit and 16-bit images are supported; bit depth is auto-detected from the first image in the sequence.

2 Installation

2.1 From Source

1. Ensure Python 3.10 or later is installed.
2. Clone or download the repository.
3. Create a virtual environment and install dependencies:

```
cd bubbletrack
python -m venv .venv

# Windows
.venv\Scripts\activate
# macOS / Linux
source .venv/bin/activate

pip install -e .
```

2.2 Running the Application

```
# Run via console entry point
bubbletrack

# Or run as a Python module
python -m bubbletrack
```

2.3 Standalone Executable

A pre-built Windows executable (`BubbleRadiusFitting.exe`) is available in the `dist/` directory. No Python installation is required. Simply double-click the executable to launch.

3 Interface Overview

The application uses a three-panel layout inspired by modern design principles.

3.1 Layout Structure

```
+-------------------------------------------------------------+
|  HEADER BAR                                     Status: Ready  |
+----------------+--+-------------------------------------------+
|  LEFT SIDEBAR  |  |  MAIN CONTENT AREA                        |
|  (collapsible) |<>|  +------------------+------------------+ |
|  - Image Source|  |  | Original Image   | Binary Mask      | |
|  - Mode Tabs   |  |  |                  |                  | |
|  - Parameters  |  |  +------------------+------------------+ |
|  - Post Proc.  |  |                                           |
|                |  |  Frame Scrubber: [<] [======o=] [>] 1/N  |
|                |  |                                           |
|                |  |  R(t) Scatter Chart                       |
+----------------+--+-------------------------------------------+
|  STATUS BAR: Frame 1/N | .tif | 3.2 um/px | ROI | Mode     |
+-------------------------------------------------------------+
```

3.2 Header Bar

The dark header bar displays the application title and a colour-coded status indicator:

Colour	Meaning	Example
Green	Success / Ready	"Ready", "R = 42.3 px"
Amber	In progress / Warning	"Fitting...", "Stopping..."
Red	Error	"Too few edge points"

3.3 Left Sidebar

The sidebar (340 px wide) contains all controls, organised into four sections:

1. **Image Source** — folder selection and file information
2. **Mode Tabs** — Pre-tune, Manual, Automatic
3. **Mode-specific Controls** — parameters and action buttons
4. **Post Processing** — export controls and physical unit settings

The sidebar can be collapsed by clicking the toggle button (</>) between the sidebar and the main content area, providing more space for the image panels.

3.4 Image Panels

Two image panels are displayed side by side:

- **Original Image** (left) — the greyscale source image with overlay graphics: green ROI rectangle, blue fitted circle, and red edge detection points.
- **Binary Mask** (right) — the processed binary image showing the detection result.

Each panel provides zoom controls in the top-right corner:

Control	Function
[+]	Zoom in (1.25×)
[−]	Zoom out (0.8×)
[]	Reset zoom to fit the view
Scroll wheel	Zoom in/out (1.15× per notch, anchored at cursor)

3.5 Frame Scrubber

The frame scrubber bar sits between the image panels and the chart. It contains:

- **Previous/Next buttons** (</>) — step one frame backward or forward.
- **Slider** — drag to jump to any frame in the sequence.
- **Frame label** — displays the current frame number and total (e.g. "Frame: 42/500").

3.6 Radius Chart

A Matplotlib scatter plot showing fitted radius (pixels) versus frame number. Data points appear as red "+" markers. The chart updates in real time during fitting operations.

3.7 Status Bar

The bottom status bar displays contextual information:

Field	Example
Frame counter	Frame: 42/500
Image format	Format: .tif
Physical scale	Scale: 3.2 µm/px
ROI bounds	ROI: [1–512, 1–512]
Current mode	Mode: Pre-tune

4 Quick Start

1. **Load images:** Click **Browse** in the Image Source section and select a folder containing image files.
2. **Navigate frames:** Use the frame scrubber slider or the `</>` buttons to browse the image sequence.
3. **Select ROI:** In the **Pre-tune** tab, click **Select ROI**, then drag a rectangle on the original image to define the region of interest.
4. **Tune parameters:** Adjust the **Threshold** and **Removing Factor** sliders until the bubble boundary appears cleanly in the binary mask. Enable advanced filters if needed.
5. **Fit a single frame:** Click **Fit Current Frame**. The fitted circle (blue) and edge points (red) appear on the original image.
6. **Batch process:** Switch to the **Automatic** tab, set the frame range, and click **Fit All**. Progress is shown in real time.
7. **Export results:** In the **Post Processing** section, set the physical parameters (FPS, scale) and click **Export Pixel Data** or **Export Physical Data** to save results as `.mat` files.

5 Analysis Modes

5.1 Pre-tune Mode

Pre-tune mode is designed for parameter optimisation on a single frame. All parameter changes update the binary mask preview in real time (debounced at 50 ms), allowing rapid visual feedback.

5.1.1 Frame Selector

A spin box allows direct navigation to any frame in the sequence (1-indexed display). This is independent of the main frame scrubber.

5.1.2 Region of Interest (ROI)

1. Click **Select ROI**. The image border turns amber to indicate selection mode.
2. Drag a rectangle on the original image panel.
3. The ROI coordinates are displayed in the sidebar and the status bar.

By default, the ROI covers the entire image. Only pixels within the ROI are used for bubble detection and circle fitting. The ROI is shown as a green rectangle on the original image.

Tip

Choose an ROI that tightly encloses the bubble. A smaller ROI improves detection accuracy and processing speed.

5.1.3 Threshold

Range	Default	Unit	Effect
0–100	50	%	Adaptive binarisation sensitivity

The threshold slider controls the sensitivity of adaptive mean thresholding. Higher values produce more white (foreground) pixels; lower values produce a stricter foreground detection. The internal sensitivity value is computed as:

$$\text{sensitivity} = \frac{\text{slider value}}{100}$$

5.1.4 Removing Factor

Range	Default	Unit	Effect
0–100	90	%	Minimum connected-component area (log scale)

This slider removes small white speckles (noise) from the binary image. The mapping is logarithmic: a slider value of 0 removes components smaller than $10\,\text{px}^2$; a slider value of 100 removes components up to 50% of the ROI area.

5.1.5 Edge Crossing Flags

Four checkboxes (Top, Right, Down, Left) indicate which edges of the ROI the bubble extends beyond. When a flag is checked, the algorithm expands the binary image in the *opposite* direction to correctly detect partial bubbles.

Note

For example, if the bubble extends beyond the top and right edges of the ROI, check **Top** and **Right**. The algorithm will expand the image downward and leftward to compensate.

5.1.6 Advanced Filters

The Advanced Filters section (initially collapsed) provides four optional image filters. Enable each filter via its checkbox.

Filter	Range	Default	Description
Gaussian Blur	1–20	3	Applied *before* binarisation. Smooths the greyscale image to suppress small noise such as tracer particles.
CLAHE	1–40	2	Applied *before* binarisation. Contrast Limited Adaptive Histogram Equalisation improves local contrast when lighting is uneven.
Morph Close	1–30	5	Applied *after* binarisation. Fills small holes and gaps inside the bubble boundary.
Morph Open	1–20	3	Applied *after* binarisation. Removes thin protrusions and spurs on the bubble edge.

5.1.7 Parameter Presets

Click **Save Preset** to store the current parameter configuration (threshold, removing factor, edge flags, and all filter settings) to a JSON file. Click **Load Preset** to restore a previously saved configuration.

5.1.8 Fit Current Frame

Click **Fit Current Frame** to run the full detection and circle-fitting pipeline on the current frame. On success:

- A blue circle is drawn on the original image showing the fitted bubble boundary.
- Red dots mark the detected edge points.
- The header displays the fitted radius in pixels.
- The $R(t)$ chart is updated.

5.2 Manual Mode

Manual mode allows you to select edge points by clicking directly on the bubble boundary.

1. Click **Select Points**. The image border turns blue and the cursor changes to a crosshair.
2. Click on the bubble edge to place points. Each click adds a red marker on the image. The point counter updates in real time.

3. A minimum of **3 points** is required. The status indicator turns green when enough points are collected.
4. Click **Done** to fit a circle through the selected points using the Taubin algorithm.
5. Click **Clear** to discard all points and start over.

Tip

For best results, distribute the selected points evenly around the bubble perimeter. Using 5–10 points typically gives reliable fits.

5.3 Automatic Mode

Automatic mode batch-processes a range of frames using the parameters configured in the Pre-tune tab.

5.3.1 Setting the Frame Range

Use the **From** and **To** spin boxes to define the processing range (1-indexed, inclusive). By default, the range covers all frames.

5.3.2 Running Batch Processing

1. Click **Fit All** to start processing. A background worker thread processes each frame sequentially.
2. The progress bar and percentage label update in real time.
3. The original image, binary mask, and $R(t)$ chart are updated periodically during processing ($\sim$150 ms intervals).
4. The fitted circle and edge points are displayed on the original image for each processed frame.

5.3.3 Stopping

Click **Stop** to request a graceful stop. The worker finishes the current frame and then halts. All results computed so far are retained.

5.3.4 Clearing Results

Click **Clear** to discard all fitting results and reset the $R(t)$ chart. This does not affect the loaded images or parameter settings.

Warning

Clicking **Clear** permanently removes all fitting results from memory. Export your data before clearing if you need to keep the results.

6 Post Processing and Export

The Post Processing section (in the lower portion of the left sidebar) provides physical unit conversion parameters and two export options.

6.1 Physical Parameters

Parameter	Range	Default	Unit	Description
FPS	0.001–99 999	10	Hz, kHz, or MHz	Frame rate of the camera. Select the unit from the dropdown (Hz, k, M).
Scale	0.001–9 999	3.2	µm/px	Spatial calibration factor.
Rmax Fit Length	3–999 (odd)	11	frames	Number of frames used for quadratic $R_{\max}$ interpolation (must be odd).

6.2 Export Pixel Data

Click **Export Pixel Data** to save raw fitting results. A file dialog opens with the default filename `radius_pixel.mat` in the image folder. The `.mat` file contains:

Variable	Size	Description
`Radius`	$(1 \times N)$	Fitted radius for each frame in pixels. Unfitted frames contain -1.
`CircleFitPar`	$(N \times 2)$	Circle centre coordinates [row, col] for each frame.
`CircleXY`	$\{1 \times N\}$	Cell array of edge-point coordinate matrices $(M_i \times 2)$ for each frame.

6.3 Export Physical Data

Click **Export Physical Data** to save time-resolved radius data in physical units. A file dialog opens with the default filename `radius_time_physical.mat`. The export process:

1. Converts pixel radii to physical units using the scale factor.
2. Converts frame indices to time using the frame rate.
3. Identifies the frame with maximum radius.
4. Fits a quadratic curve $R(t) = at^2 + bt + c$ around the peak using the specified fit window.
5. If the parabola is concave ($a < 0$), computes the sub-frame peak: $t_{\text{peak}} = -b/(2a)$, $R_{\text{peak}} = (4ac - b^2)/(4a)$.
6. Shifts the time axis so $t = 0$ at the peak radius location.

The `.mat` file contains:

Variable	Size	Description
`t`	$(N{+}1 \times 1)$	Time array in seconds, centred at $R_{\max}$.
`R`	$(N{+}1 \times 1)$	Radius array in µm.
`RmaxAll`	scalar	Peak radius value in µm.
`RmaxTimeLoc`	scalar	Time of peak radius in seconds.

Note

If the $R_{\max}$ fit window extends beyond the data boundaries, the export will fail with an error message. Reduce the **Rmax Fit Length** parameter or ensure the peak radius is sufficiently far from the edges of the frame range.

7 Algorithm Overview

This section provides a technical description of the image processing and circle-fitting pipeline.

7.1 Processing Pipeline

The detection pipeline consists of the following steps, executed in order:

1. **Load and normalise** — Read the raw image and normalise pixel values to the $[0, 1]$ range based on the detected bit depth.
2. **Gaussian Blur** (optional) — Apply a Gaussian smoothing kernel with the specified σ to suppress high-frequency noise. Applied *before* binarisation.
3. **CLAHE** (optional) — Apply Contrast Limited Adaptive Histogram Equalisation with the specified clip limit to improve local contrast. Applied *before* binarisation.
4. **Adaptive thresholding** — Convert the greyscale image to binary using local mean thresholding (OpenCV `adaptiveThreshold`). The block size is computed automatically as:

$$B = 2\left\lceil \frac{\min(H, W)}{16} \right\rceil + 1 \qquad (B \geq 3)$$

   The threshold offset is derived from the sensitivity parameter:

$$C = (\text{sensitivity} - 0.5) \times 255$$

5. **ROI extraction** — Crop the binary image to the user-defined region of interest.
6. **Boundary expansion** — If the bubble extends beyond the ROI, the image is doubled along the non-crossing edge direction and filled with white pixels to complete partial bubbles.
7. **Small object removal** — Connected components with area below the removing factor threshold are removed.
8. **Morphological opening** (optional) — Erosion followed by dilation with a disk structuring element to remove thin protrusions.

9. **Morphological closing** (optional) — Dilation followed by erosion with a disk structuring element, then hole filling, to close gaps in the bubble boundary.
10. **Inversion** — The image is inverted so that the bubble interior (dark in the original) becomes the foreground.
11. **Blob filtering** — Connected components are labelled and filtered by shape:
    - Reject if major axis $> 2.2\times$ short image side
    - Reject if eccentricity > 1.6 (major/minor axis ratio)
    - Select the largest remaining blob
12. **Edge detection** — Extract the boundary using a morphological gradient: dilate the blob mask with a 3×3 kernel, then XOR with the original mask.
13. **Crop back** — If the image was expanded in step 6, crop back to the original ROI dimensions.

7.2 Circle Fitting: Taubin Method

The application uses the Taubin algebraic method for circle fitting, which provides a robust least-squares estimate that is invariant to rotation and translation.

Given $N \geq 3$ boundary points $\{(x_i, y_i)\}$, the algorithm:

1. Computes the centroid $(\bar{x}, \bar{y})$ and centres the data.
2. Builds a covariance-based system from the centred coordinates.
3. Solves a characteristic polynomial via Newton's method (up to 20 iterations, tolerance 10^{-12}).
4. Extracts the circle centre (x_c, y_c) and radius r from the solution.

Reference

G. Taubin, "Estimation Of Planar Curves, Surfaces And Nonplanar Space Curves Defined By Implicit Equations, With Applications To Edge And Range Image Segmentation," *IEEE Trans. PAMI*, Vol. 13, pp. 1115–1138, 1991.

7.3 Radius Outlier Rejection

After circle fitting, the fitted radius is compared against the maximum image dimension (the longer side of the image). If the radius exceeds this value or is `NaN`, the result is rejected and the frame is marked as unfitted.

7.4 $R_{\max}$ Quadratic Interpolation

During physical data export, the software identifies the maximum radius and fits a parabola to the surrounding data points:

$$R(t) = at^2 + bt + c \tag{1}$$

If the fit yields a concave parabola ($a < 0$), the sub-frame peak is computed analytically:

$$t_{\text{peak}} = -\frac{b}{2a} \tag{2}$$

$$R_{\text{peak}} = \frac{4ac - b^2}{4a} \tag{3}$$

This provides sub-frame temporal precision for determining the maximum bubble radius.

8 Parameter Reference

8.1 Image Processing Parameters

Parameter	Range	Default	Unit	Effect
Threshold	0–100	50	%	Adaptive binarisation sensitivity. Higher = more foreground pixels.
Removing Factor	0–100	90	%	Minimum component area (logarithmic scale). Higher = removes larger noise.
Gaussian σ	1–20	3	px	Pre-binarisation smoothing strength.
CLAHE Clip	1–40	2	—	Contrast enhancement strength.
Morph Close	1–30	5	px	Post-binarisation gap-filling radius.
Morph Open	1–20	3	px	Post-binarisation spur-removal radius.
Edge Flags	4 bools	all off	—	Bubble crosses [Top, Right, Down, Left] edges.

8.2 Physical Conversion Parameters

Parameter	Range	Default	Unit	Effect
FPS	0.001–99 999	10	Hz/kHz/MHz	Camera frame rate
Scale	0.001–9 999	3.2	µm/px	Pixel calibration
Rmax Fit Length	3–999	11	frames	Quadratic fit window (odd)

8.3 Internal Constraints

Constraint	Value	Description
Min. edge points	3	Required for circle fitting
Max. radius	$\max(H, W)$	Image long-side dimension
Min. hole area	$40\,\text{px}^2$	Holes smaller than this are filled
Max. axis ratio	2.2	Blob major axis / short image side
Max. eccentricity	1.6	Major / minor axis ratio

9 Typical Workflows

9.1 Workflow 1: Single-Frame Analysis

This workflow is useful for quickly inspecting individual frames or verifying parameter settings.

1. Load the image folder via **Browse**.
2. Navigate to the frame of interest using the frame scrubber.
3. In the **Pre-tune** tab:
   a. Select an ROI around the bubble.
   b. Adjust **Threshold** until the bubble interior is clearly segmented.
   c. Increase **Removing Factor** to eliminate noise spots.
   d. Enable **Gaussian Blur** if small particles (e.g. tracer beads) are present.
   e. Enable **CLAHE** if lighting is uneven.
   f. Check edge flags if the bubble is partially outside the ROI.
4. Click **Fit Current Frame** and inspect the result.
5. Optionally save the parameter preset for later use.

9.2 Workflow 2: Batch Processing

This is the primary workflow for extracting $R(t)$ data from a complete image sequence.

1. Complete the parameter tuning (Workflow 1) on a representative frame.
2. Switch to the **Automatic** tab.
3. Set the **From** and **To** frame numbers.
4. Click **Fit All**. Monitor progress via the progress bar and the $R(t)$ chart.
5. Once complete, review the chart for outliers or missing data points.
6. In **Post Processing**:
   a. Set the camera **FPS** and spatial **Scale**.
   b. Click **Export Pixel Data** for raw results.
   c. Click **Export Physical Data** for time-resolved physical-unit results with $R_{\max}$ interpolation.

9.3 Workflow 3: Manual Point Selection

Use this workflow when the automatic detection fails to identify the bubble boundary correctly (e.g. due to poor contrast or complex shapes).

1. Navigate to the target frame.
2. Switch to the **Manual** tab.
3. Click **Select Points** and click 5–10 points around the bubble edge.
4. Click **Done** to fit the circle.
5. Repeat for additional frames if needed.
6. Export results as in Workflow 2.

10 Troubleshooting

Problem	Possible Cause	Solution
"No images found" after selecting a folder	Folder contains no supported image files	Ensure the folder contains `.tiff`, `.tif`, `.png`, `.jpg`, or `.bmp` files.
Binary mask is mostly white or mostly black	Threshold too high or too low	Adjust the Threshold slider until the bubble boundary is clearly visible.
Small noise spots remain in the binary mask	Removing Factor too low	Increase the Removing Factor slider. Enable Gaussian Blur to suppress fine noise.
"Too few edge points" after fitting	Bubble not detected or ROI too small	Enlarge the ROI, adjust the Threshold, or check the edge flags.
"Radius outlier, skipped"	Fitted circle is unreasonably large	The detection picked up the wrong region. Adjust Threshold, Removing Factor, or use Morph Open/Close to clean the boundary.
Bubble boundary has gaps	Weak contrast at some edges	Enable **CLAHE** to improve local contrast, or enable **Morph Close** to fill gaps.
Bubble boundary has spurs or protrusions	Noise connected to the boundary	Enable **Morph Open** to remove thin protrusions.
Stop button does not respond immediately	Worker finishes current frame before stopping	This is expected. The stop is graceful—wait a moment for the current frame to complete.
Export fails with "Rmax_Fit_Length exceeds data range"	Fit window is wider than available data around the peak	Reduce the **Rmax Fit Length** parameter, or process more frames around the peak radius.
Application window is too small	Display scaling or resolution issue	The window can be freely resized. Collapse the left sidebar for more image space.

11 Keyboard and Mouse Reference

Mode	Input	Action	Result
Any	Scroll wheel (on image)	Zoom in/out	1.15× per notch
Any	Click `[+]`/`[-]`/`[]`	Zoom/Reset	Button-based zoom control
Normal	Drag on image	Pan	Scroll hand drag
Point	Left-click on image	Place point	Red dot added
ROI	Left-drag on image	Draw rectangle	ROI defined
Any	Click `</>`	Step frame	±1 frame
Any	Drag frame slider	Jump to frame	Instant navigation

12 Export File Format Details

All exported files use the MATLAB `.mat` (v5) format and can be loaded in MATLAB using the `load` function or in Python using `scipy.io.loadmat`.

12.1 Pixel Data (`radius_pixel.mat`)

```
data = load('radius_pixel.mat');

% Radius array (1 x N), in pixels. -1 = unfitted frame.
R = data.Radius;

% Circle centres (N x 2), [row, col] for each frame.
centres = data.CircleFitPar;

% Edge points: cell array {1 x N}.
% Each cell contains an (M x 2) matrix of [row, col] coordinates.
edges = data.CircleXY;
```

12.2 Physical Data (`radius_time_physical.mat`)

```
data = load('radius_time_physical.mat');

% Time array (N+1 x 1), in seconds. Centred at Rmax.
t = data.t;

% Radius array (N+1 x 1), in micrometres.
R = data.R;

% Maximum radius (scalar), in micrometres.
Rmax = data.RmaxAll;

% Time of maximum radius (scalar), in seconds.
t_Rmax = data.RmaxTimeLoc;
```

```
% Plot R(t)
figure;
plot(t * 1e6, R, 'r.');  % convert to microseconds
xlabel('Time␣(\mus)');
ylabel('Radius␣(\mum)');
title(sprintf('R_{max}␣=␣%.1f␣\\mum', Rmax));
```

Acknowledgements

This software was developed at The University of Texas at Austin. The circle fitting algorithm is based on the Taubin method as described in:

> G. Taubin, "Estimation Of Planar Curves, Surfaces And Nonplanar Space Curves Defined By Implicit Equations, With Applications To Edge And Range Image Segmentation," *IEEE Trans. PAMI*, Vol. 13, pp. 1115–1138, 1991.